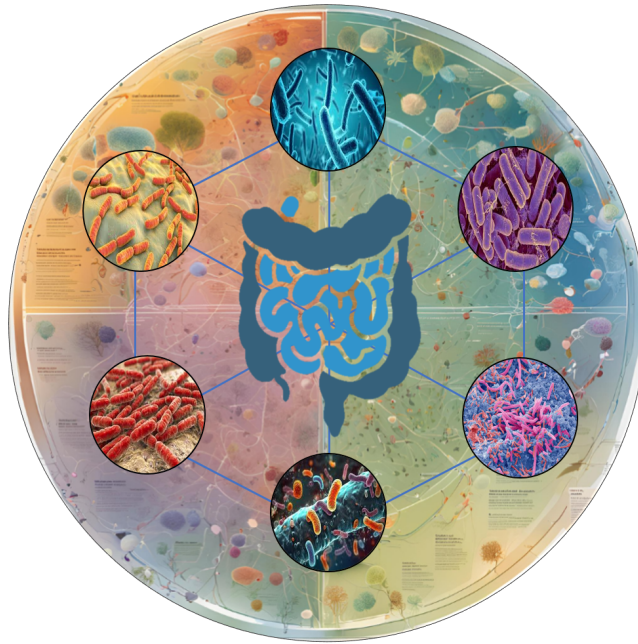




Gut Microbiome Report

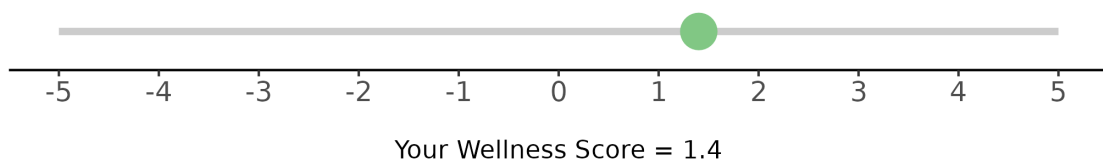
Lab ID: 0BPXYV

Date of sample collection: 4/5/2024

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Gut Microbiome Wellness Index 2 (GMWI2):

GMWI2 is a standardized disease-agnostic health status indicator based on gut microbiome taxonomic profiles. A positive GMWI2 value indicates a greater abundance of healthy, beneficial taxa, while a negative GMWI2 value reflects a higher abundance of less beneficial taxa. When the GMWI2 value is 0, it shows a balanced presence of beneficial and less beneficial taxa, categorizing the sample as neutral in terms of health status. ([Chang et al., "Gut Microbiome Wellness Index 2 enhances health status prediction from gut microbiome taxonomic profiles", *Nature Communications*, 2024, Vol. 15, Issue 1, pp. 7447](#))



Your gut microbiome wellness score is 1.4 which indicates good gut microbiome.

Interpretation of Your Results:

Your overall gut microbiome analysis indicates a well-balanced and healthy microbiome, supporting optimal gut health.

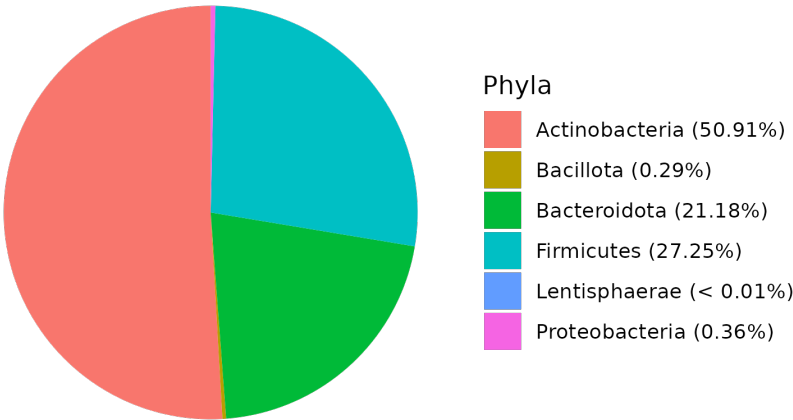
Detailed Gut Microbiome Analysis

This report is the analysis of the bacterial diversity and abundance of different species present in the gut microbiome using stool sample for metagenomic profiling. Through this report we aim to offer valuable insights, empowering you to gain a deeper understanding of your gut health. From the stool sample collected from you by the cohort team, we isolated metagenomic DNA, which is sequenced using Next-generation sequencing (NGS) technology. The sequence data was analyzed using different bioinformatics pipelines to profile the microbial community in the sample. This information was used to generate your gut microbiome report. We aspire to equip you with insights that can guide you towards making informed decisions about your health and lifestyle. We have reported your results in the following sections:

1. Phylum Diversity
2. Firmicutes / Bacteroidetes Ratio
3. Species-level Diversity
 - 3.1 Species Richness
 - 3.2 Evenness Index
 - 3.3 Microbiome Diversity Index
 - 3.4 Top 15 Species
4. Probiotic Species and Short Chain Fatty Acid (SCFA) Producers

Phylum-Level Diversity of Your Gut Microbiome:

The phylum diversity represents the variety of bacterial phyla present in your gut microbiome. Bacterial phyla are broad categories that contain different groups of bacteria sharing certain characteristics. In your microbiome report, we analyzed the number and relative abundance of various bacterial phyla present in your gut.

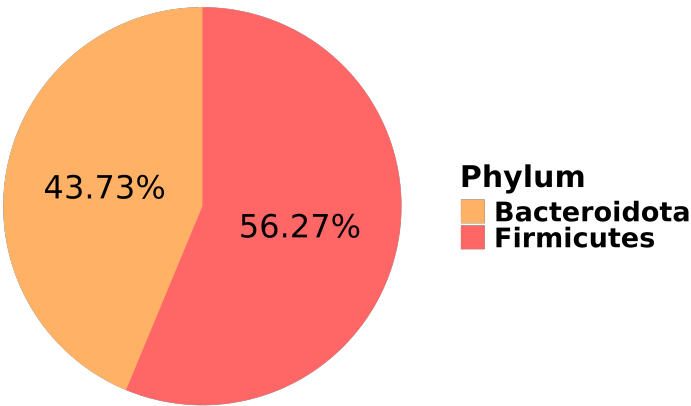


Your gut microbiome profile indicates the presence of 6 distinct phyla. The average range of phyla present in gut microbiome of Indian population is 3-15.*

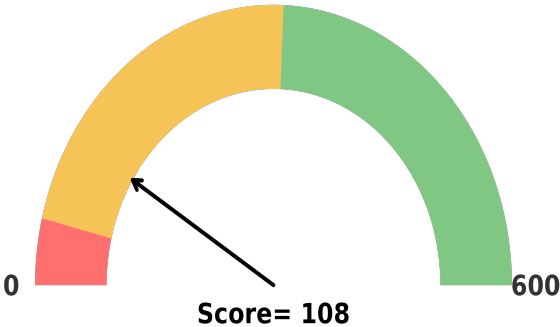
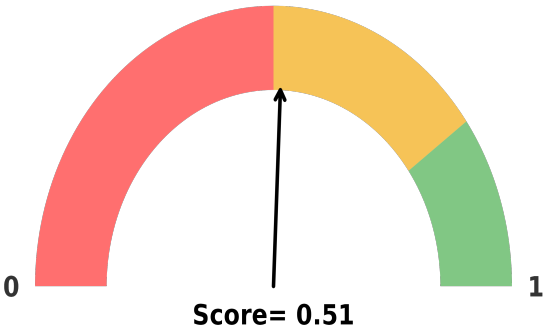
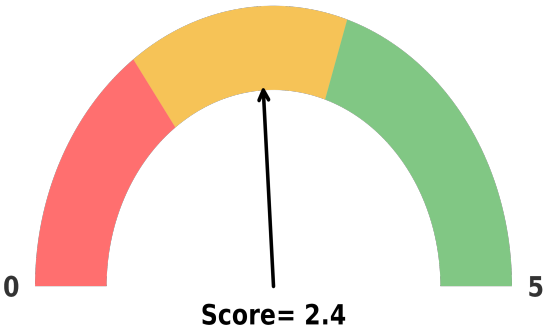
Firmicutes and Bacteroidetes Ratio:

Firmicutes (F) and Bacteroidetes (B) are the most common bacterial phyla found in the gut microbiome. The ratio between the abundance of Firmicutes and Bacteroidetes, known as the F/B ratio, is used to measure the relative proportion of these organisms. A balanced proportion (F/B ratio = 1) of these two phyla is vital for maintaining good health, while the majority of the Indian population typically falls within an F/B ratio range of approximately 0.69 to 4.41.*

F / B = 1.29



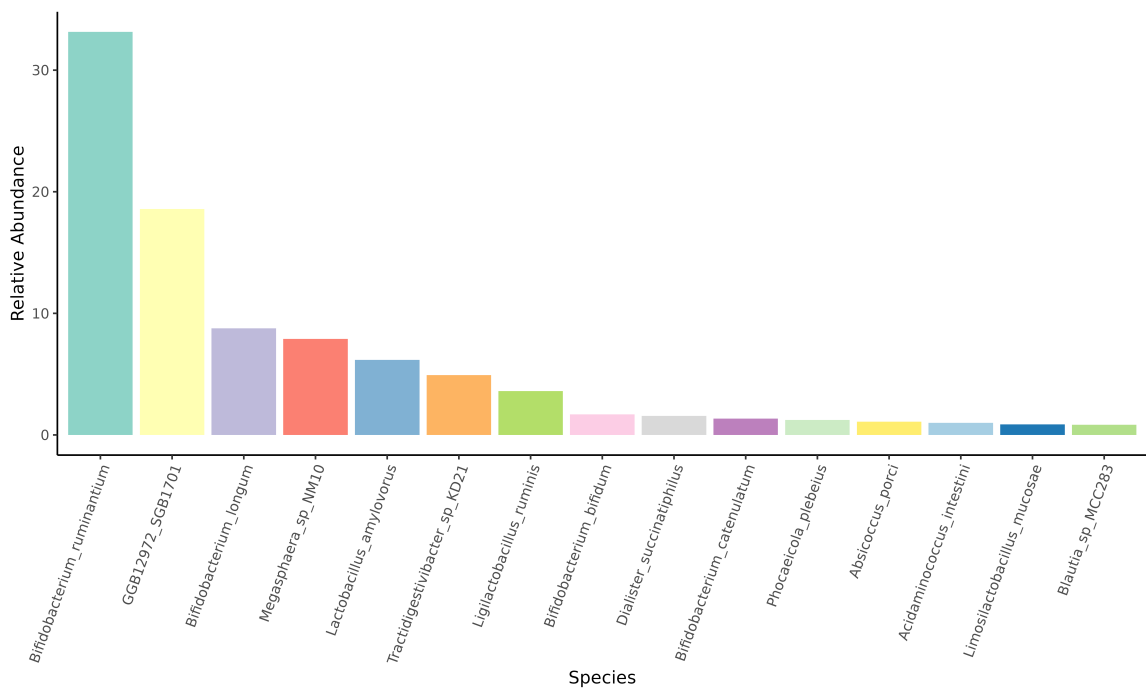
Species-level Diversity of Your Gut Microbiome*:

Microbial Species Richness	Description
 Score= 108	Richness is the total number of microbial species identified within your sample. The number of species found in your gut microbiome is 108 and it is within the average Indian population range.
Evenness Index	Description
 Score= 0.51	Evenness describes how uniformly all the species are present in your gut microbiome. The value of species evenness in your sample is 0.51 and it is within the average Indian population range.
Microbiome Diversity Index	Description
 Score= 2.4	The microbiome diversity index is calculated using both richness and evenness. The value of microbiome index of your sample is 2.40 and it is within the average Indian population range.

Distribution of Species:

The top 15 species in your gut microbiome based on the relative abundance.*

*When a bacterial species has numbers in its name, it signifies a recent discovery that has not been formally named yet.



Probiotic Species:

Probiotic species are microorganisms present in your gut microbiome that contribute in maintaining a good health supporting digestion, modulating the immune system, protecting the intestinal barrier and reducing inflammation. They also make the gut a hospitable place for the survival of different types of microorganisms. Some of these organisms are also known producers of Short Chain Fatty Acids (SCFAs). SCFAs are fatty acids that are produced by the gut microbiota during the fermentation of dietary fibers. SCFAs improve gut barrier function, possess anti-inflammatory and immunomodulatory properties, which help promote overall gut health and lower the risk of gastrointestinal diseases. (Xiong et al., "Health benefits and side effects of short-chain fatty acids", *Foods*, 2022, Vol. 11, Issue 18, pp. 2863)

This section presents the probiotic species detected in your gut microbiome and SCFA produced by them.

Species	Presence	Short Chain Fatty Acid/s Produced	Description
<i>Alistipes putredinis</i>	+	Acetate, Propionate, Butyrate	It is a bacteria from the phylum Bacteroidetes. It has been shown to have anti-inflammatory properties.
<i>Anaerostipes hadrus</i>	-	Not available	Not available
<i>Bifidobacterium adolescentis</i>	-	Acetate, Propionate, Lactate	It is shown to have anti-inflammatory and anti-oxidant properties. It is also used as a probiotic to restore gut health.

<i>Bifidobacterium bifidum</i>	+	Acetate, Lactate	It is a bacteria known for their ability to ferment carbohydrates, help with digestion, nutrient absorption, and fighting off "bad" organisms that can cause disease.
<i>Bifidobacterium breve</i>	+	Acetate, Lactate	It is shown to have antimicrobial activity against human pathogens and immuno-stimulating properties.
<i>Bifidobacterium catenulatum</i>	+	Acetate	It is shown to inhibit harmful pathogens, contribute to the maintenance of a balanced microbiome and support overall gut health and immune function.
<i>Bifidobacterium longum</i>	+	Acetate, Lactate	It is shown to have anti-inflammatory properties, plays role in immunoregulation, and decreasing the infection risk. It is also used as a commercial probiotic to restore gut health.
<i>Blautia wexlerae</i>	+	Not available	It is shown to alter energy metabolism of the host and anti-inflammatory properties. It is also shown to change the composition of the gut microbiome.
<i>Collinsella aerofaciens</i>	-	Not available	Not available
<i>Coprococcus catus</i>	+	Butyrate, Acetate	It is a bacteria that plays an important role in fermenting dietary fibers and producing propionate, butyrate and supports gut health.
<i>Dorea longicatena</i>	-	Not available	Not available
<i>Faecalibacterium prausnitzii</i>	+	Butyrate, Acetate	It is one of the most abundant bacterial species found in the gut. It is known for it's anti-inflammatory properties and importance in gut health.
<i>Fusicatenibacter saccharivorans</i>	+	Lactate, Acetate, Succinate	It is bacterial species shown to have the ability to reduce inflammation, improve colitis symptoms, and restore gut microbiota balance in patients with ulcerative colitis.
<i>Lactobacillus ruminis</i>	-	Not available	It inhibits harmful pathogens, enhances carbohydrate utilization for energy, and helps maintain a balanced microbial community, thereby supporting digestion and overall immune function.
<i>Megasphaera elsdenii</i>	-	Not available	Not available

<i>Prevotella copri</i>	-	Butyrate	It produces short-chain fatty acids (SCFAs) that protect the mucosal barrier and reduce inflammation. It has been associated with reduction of visceral fat and improved glucose metabolism.
<i>Roseburia faecis</i>	+	Butyrate	It is shown to modulate immune response, reduce inflammation, improve symptoms of irritable bowel syndrome (IBS), and enhance the overall composition of the gut microbiota.
<i>Roseburia hominis</i>	-	Butyrate	It is a bacteria known to produce beneficial short-chain fatty acids (SCFAs) and it has been shown to reduce neuroinflammation.
<i>Roseburia intestinalis</i>	-	Butyrate	It is shown to prevent intestinal inflammation and maintain energy homeostasis. It is also considered as a potential probiotic for treating gut associated disorders.
<i>Ruminococcus bromii</i>	+	Acetate, Propionate, Butyrate	It is bacteria which breaks down complex carbohydrates, producing short-chain fatty acids important for colon health. It also helps promote gut microbiome balance.

Among the Top 16 probiotic species predominantly observed in the Indian population, we identified 11 species in your gut microbiome.*

***Note:** The reference ranges for the interpretation of the results are based on the analysis of more than 2500 gut microbiome samples collected across India in Phenome India: CSIR Cohort Health Knowledgebase.

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Contact us at: microbiome.igib@gmail.com

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